

attribute (h_k) from the last layer of the RNN, for each k .

- h_n : This is a tensor of size ($num_layers * num_directions, batch, hidden_size$). It contains the hidden state for $k = seq_len$.

As stated earlier, both `torch.nn.GRU` and `torch.nn.LSTM` have the same API, i.e., they accept the same set of parameters and accept inputs in the same format, even returning them in the same format too.

torch.nn.RNNCell

After all, this represents only a single cell of the RNN, so it accepts only four parameters — all of which have the same meaning as they did in `torch.nn.RNN`.

Parameters

- `input_size`: The number of expected features in the input x .
- `hidden_size`: The number of features in the hidden state h .
- `bias`: If 'False', then the layer does not use bias weights b_{ih} and b_{hh} . Default: True.
- `non-linearity`: This is the non-linearity to use. It can be either `tanh` or `relu`. Default: `tanh`.

Again, since this is just a single cell of an RNN, the input and output dimensions are much simpler.

Inputs

- `input`: This is a tensor of shape ($batch, input_size$) that contains the input features.

- `hidden`: This is a tensor of shape ($batch, hidden_size$) that contains the initial hidden states for each of the elements in the batch.

Output

h : This is a tensor of shape ($batch, hidden_size$) and it gives us the hidden state for the next time step. **END** 

References

- [1] <https://en.wikipedia.org/>
- [2] <https://pytorch.org>

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IoT

At present, millions of sensors are being added to the IoT ecosystem every year. The proliferation of IoT devices is happening at a very fast pace, posing a serious data challenge. In addition, the output value of these sensors is used for various purposes. Be it smart home, smart energy meter, or smart manufacturing hub, if this data is not accurate and not validated properly, then this may lead to a disastrous situation. Even minute variations in this data can lead to an unwanted output or outcome.

The decisions taken based on this data need to be accurate, as they will influence later outcomes. Sometimes sensor devices produce wrong results for a short period of time due to unnatural events like insects covering the sensors at home, or a leaf or some object covering sensors installed outdoors. When multiple IoT devices are involved, selecting the sequence of devices to perform certain tasks is a challenge.

Various ML or DL algorithms and techniques are used to solve the above mentioned challenges.

- The output data collected during this threshold time is time series data,

which is fed to the LSTM or ARIMA model to detect anomalies.

- *KNN approach*: With this approach the output data of sensor devices is verified with K-Nearest sensor devices. The distance between two sensor devices is measured using the location (x,y) values of both devices. This helps to classify the sensors in the correct group, and to categorise them properly.

- The combination of time series data and the output of various sensors can be used for identifying and distributing tasks to appropriate sensors.

This article can be used as reference material for building any AI, ML or DL model across industry sectors, in combination with other emerging technologies like IoT, AR/VR, etc. **END** 

References

- [1] <https://www.tensorflow.org/>
- [2] <https://keras.io/>
- [3] <https://opensourceforu.com/2017/12/deep-learning-using-algorithms-to-make-machines-think/>
- [4] <https://www.vox.com/future-perfect/2020/2/14/21063487/self-driving-cars-autonomous-vehicles-waymo-cruise-uber>
- [5] <https://www.forbes.com/sites/bernardmarr/2019/12/20/what-is-the-artificial-intelligence-of-things-when-ai-meets-iot/#4300af1eb1fd>

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